NORDDESIGN 2026 CONFERENCE
TAMPERE, FINLAND, 12-14 AUGUST 2026

Early Exploration and Benchmarking of Soft Gripper Designs for Robotic Grasping of Fish

Benjamin Karlsen[1], Øystein Bjelland[2], Agus Hasan[1], Lars André Giske[3]
[1]Department of ICT and Natural Sciences, NTNU – Norwegian University of Science and Technology
[2]Department of Ocean Operations and Civil Engineering, NTNU – Norwegian University of Science and Technology
[3]Optimar AS

Abstract: This paper presents exploration of early-stage gripper designs for robotic grasping of fish and evaluate prototype performance through load capacity benchmark tests. The research is motivated by the need for automation within the Norwegian fishing industry, and designing for Physical-AI based control methods. We explore several gripping technologies, including compliant fingers, pneumatic soft robotic fingers, standard suction, Bernoulli suction and freezing, and present four hybrid-design prototypes combining suction with compliant and pneumatic soft robotic fingers. We find that a hybrid design with suction and compliant fingers meet industrial design requirements by grasping a polyurethane fish model with a normal vertical load capacity of 215 N, and a lateral horizontal load capacity of 65 N.

Keywords: Prototyping, Design-for-AI, Iterative design, Robotic grasping, Maritime mechatronics

Introduction

The fishing industry is Norway's second largest export industry, exporting a total of 2.8 million tons of seafood with a value of around 16.1 billion Euro in 2025 (Hatlem, 2026). In recent years, much effort has focused on automating the Norwegian fishing industry, with a particular focus on automating single processes. In a feasibility study by Giske et al., (2025a), it was concluded that combining different solutions to automate the whole production flow is possible, given that the fish is singulated. We use the term singulate to describe the process of isolating and orienting individual fish from a batch. Preliminary work has been conducted where different singulation concepts have been simulated in a physics engine (Giske et al., 2025b). This work demonstrated that rotation-based mechanical solutions were feasible for performing singulation tasks. However, the proposed solutions require much space, which can be scarce in land-based facilities and on-board fishing trawler vessels. Robotic solutions on the other hand are space efficient, but robust grasping and manipulation of soft biological objects is still a challenge (Khan et al., 2018). Little research has explored the potential of artificial intelligence (AI) based control for robotic grippers in the context of marine seafood processing. Soft robotic grippers with physical AI-based control algorithms hold the potential of performing singulation tasks of fish specimens in dynamic marine environments. For example, recent studies demonstrated grasping compliant objects with robot imitation learning from a human supervisor in virtual reality (Dyrstad et al., 2018; Dyrstad and Mathiassen, 2017; Misimi et al., 2018). Because soft robot actuators are normally highly compliant, nonlinear, may exhibit hysteresis, and are high-dimensional (continuous), controllers depend on stable and repeatable dynamics, observability and low-latency actuation for precision. Grasping fish is a delicate balance as excessive grasping forces could compromise product quality, while too low force could lead to manipulation failure. The fish also has variable compliance. Investigating the unknown physical limitations of gripping whole fish is a necessary foundation for robust robotic manipulation, including the training of physical-AI based control. Therefore, the objective of this study is to iteratively investigate and benchmark gripper designs for grasping whole fish specimens, with particular emphasis on evaluating their load-bearing capacity.

Related Work

An unstructured literature review was initiated with search words "fish+gripper+robot" and "fish+grasp+robot" to investigate previous work. While the technology for handling fish filets is quite mature (Kvam et al., 2023; Phan, 2025), fewer studies were identified on handling whole round fish or similar objects.

Some studies have focused on sophisticated control with two- or three-finger robotic grippers. One study used a two-finger soft gripper with robot imitation learning for picking Atlantic herring from a bulk of 25 fish (Dyrstad et al., 2018; Dyrstad and Mathiassen, 2017). They achieved a success rate of 74% and reported that the two most common failure types were that (i) the robot failed to pick up the fish, and (ii) the fish was dropped during transfer. Misimi et al., (2018) used a three-finger flexible tactile robot hand (ReFlex TakkTile, RightHandRobotics, USA) for grasping compliant objects. Similarly, they achieved a 75% success rate. Danielsen and Winsjansen (2019) developed a rigid dexterous gripper for general food handling, but applications were not demonstrated. Reite (2023) developed a rigid two-finger concept, but did not demonstrate feasibility on real fish. Fang et al (2023) demonstrated dynamic grasping with a parallel gripper for swimming robot fish in water. They extended the fingertips of a Robotiq-85 gripper by mounting two extra 3D-printed parallel jaws. They achieved a success rate of 75.5% and highlighted low friction as the main cause of failure in 46% of the cases, and poor grasp quality in about 15% of the cases. From these studies, it appears that two- or three finger grippers are not sufficient on their own.

Other studies have investigated suction cup-based handling of whole fish, for example Espelund (2022). Here, mackerel, a small and stiff fish with a little variation in size was studied. This contrasts cod and haddock, which are common species on the Norwegian coast and have a much larger variation in size and stiffness. Another study by Giske et al (2025b) have explored suction cup designs to lift and drag cod. The main problem observed was that varying size and shape caused the suction cup to not completely seal, concluding that further physical tests on gripper design was necessary. In addition, defining a well-posed grasping point was mentioned as a major challenge. Given that the suction cup was not enough on its own, it could suggest the need to combine existing solutions to handle different parts of the grasp.

On gripper systems for general food handling, Low et al. (2022) presented a soft robotic gripper system consisting of three fingers which could be reconfigured. They concluded that although reconfigurability improved gripping function, it also increased operational complexity. Wang et al. (2021) used a scooping plate with rubber strings for grabbing various food products. Caldwell (2013) summarized and reviewed eight different robotic gripper technologies for grasping food. Table 1 shows performance parameters for these technologies, including two- or three rigid fingers, enclosing, short needles, long needles, standard suction, Conda suction, Bernoulli suction and freezing. The parameters seen as most critical for fish handling have been highlighted, and include the ability to handle uneven surfaces, irregular shapes, soft structures, fragile objects, moist surfaces, and hygienic performance. In this context, fish fragility is defined as the force threshold at which tissue damage occurs.

Gripper Type	Hard surface	Uneven surface	Irregular shape	Soft structure	Porous	Fragile	Precise positioning	Single side grip	Dry surface	Moist surface	Heavy object	Grip marks	Hygienic performance
2 / 3 rigid fingers	+	+/0	0/-	0/-	0	-	+	-	+	+	+	0/-	0/-
Enclosing object	+	+	+	+	+	+	-	-	+	+	+	+	0/-
Short needles	-	-	-	+	+	-	+	+	+	+	-	-	-
Long needles	-	+	+	+	+	-	+	+	+	+	+	-	0/-
Standard suction	+	-	+	+/0	-	+	+	+	+	-	+		-
Conda suction	+	+/0	+	+	+	+	+	+	+	+	-	+	+/0
Bernoulli suction	+	-	+	+	-	+	-	+	+	+	+	+	+
Freeze grip	+/0	+/0	+	+	+	+	+	+	-	+	+	+	+

Table 1. Qualitative performance parameters for gripper technologies, as presented in Caldwell (2013). + indicates good performance, 0 indicates neutral, and – indicates poor performance. The green areas are identified as important for handling fish.

In recent years, soft robotic technologies such as soft fingered grippers have been researched. Terrile et al. (2021) explored four different gripper technologies; a four-finger silicone soft finger, a granular jamming gripper, a three-finger articulated rigid body gripper, and compliant three-finger design. These were tested by grasping 12 different objects in normal, humid, and dusty environments. They found that the compliant and flexible finger grippers outperformed the articulated rigid gripper and granular jamming gripper. Table 2 shows the results of flexible fingers, articulated rigid fingers, granular jamming, and compliant fingers.

Object	**Flexible finger average**	**Rigid finger average**	**Granular Jamming average**	**Compliant Fingers average**
Tennis ball	100%	42%	0%	100%
Screen cleaner	100%	75%	0%	100%
Vaseline container	67%	0%	0%	100%
Metallic container	87%	0%	0%	58%
Nut	100%	0%	73%	100%
Rock	17%	0%	58%	83%
Tomato	65%	93%	0%	100%
Egg	0%	0%	0%	100%
Glue stick	100%	20%	98%	100%
Dish sponge	85%	57%	0%	93%
Broccoli	97%	68%	0%	100%
Candy box	100%	0%	98%	3%
Overall average	**77%**	**30%**	**27%**	**86%**

Table 2: Average results from testing flexible gripper technologies on different objects in three environments Terrile et al., (2021).

From Caldwell (2013) and Terrile et al (2021), we have created a morphology table to summarize the most relevant performance metrics, and therefore the solution space for fish grasping (see Table 3). The results from the soft gripper tests in Table 2 have been transformed into the same grading as in Table 1. Additionally, the needle grippers have been removed, as they introduce unwanted punctures in the fish which could cause contamination. Dini et al (2009) supports

this by highlighting enclosure, vacuum, Bernoulli suction, freeze gripping and needle as the most common principles for grasping food.

Based on the literature review, we hypothesize that multiple grasping technologies must be combined to meet industrial requirements for fish grasping and load capacities. Our research goal is therefore to explore and benchmark robot gripper designs based on (i) isolated grasping technologies, and (ii) hybrid grasping designs. The contributions of the paper are novel gripper designs for grasping whole fish, and quantifying the load capacity of these gripper designs and technologies. The remainder of the paper is structures as follows. First we present our design approach and test setup for measuring gripper load capacity. Then, we present prototyping of designs based on five gripping technologies. We then present test results with respect to industrial requirements. Finally, some discussions and concluding remarks are provided.

	Rigid fingers	Enclosing	Standard suction	Conda suction	Bernoulli suction	Freeze	Soft fingers	Compliant fingers	Granular jamming
Uneven surface	+/0	+	-	+/0	-	+/0	+	+	-
Irregular shape	0/-	+	+	+	+	+	+	+	-
Soft structure	0/-	+	+/0	+	+	+	+	+	0
Moist surface	+	+	-	+	+	+	+	+	+
Fragile	-	+	+	+	+	+	+	+	-
Hygienic	0/-	0/-	-	+/0	+	+	0	+	0

Table 3: Morphology table from gripper technologies based on Caldwell, (2013) and Terrile et al., (2021). + indicates good performance, 0 indicates neutral, and – indicates poor performance.

Method

Design Approach

Using the morphology table as a baseline, multiple gripper designs could meet the initial design requirements by combining different technologies. To test concepts outlined in the morphology table, the design approach consisted of quick iteration cycles of design, prototyping, and benchmarking, following the Wayfaring method by Steinert and Leifer, (2012).

To establish initial design requirements and benchmark gripper designs, we investigated datasheets of known fish processing machines. Parameters from a Baader 444 (Baader Global SE, Lübeck, Germany) was chosen for benchmark as it is widely used in the industry and is likely to be fed with singulated fish. The working range of the Baader 444 is 40-90 cm long fish with a weight of 0.5-11 kg depending on the fish species and machine model. Additionally, the machine has a throughput of 15-35 fish/minute. From these specifications, the edge-case scenario of the gripper can be defined as 35 fish at 90 cm and 11 kg a minute.

Benchmark Test Setup

To evaluate developed prototype gripper designs with respect to load capacity, a benchmark test setup was created. The gripper prototypes were mounted to an Omron TM5 industrial robot arm (Omron, Kyoto, Japan), and the task was to grasp and lift a moulded fish model. The moulded fish model was made from Xencast PX30 shore A 30-35 resin (Easy Composites, UK), inserted with sponges to increase the deformability (see Figure 1A). The fish model was coated in green soap to mimic mucus. Weights were added to the fish model incrementally until gripping was unsuccessful. Additionally, a horizontal force was applied to the fish model along the red axis in Figure 1B to see what it would take for the fish to slide in its least resistant direction inside the gripper. Forces were measured using a digital tensile load cell (Biltema, 48-825) with a capacity of about 3000 N and accuracy of about 1 N. The vertical and horizontal grasp limits were compared to the specifications of the Baader 444. As benchmark tests were done consecutively after each design iteration, results informed insights which were used as a baseline in the next iteration cycle.

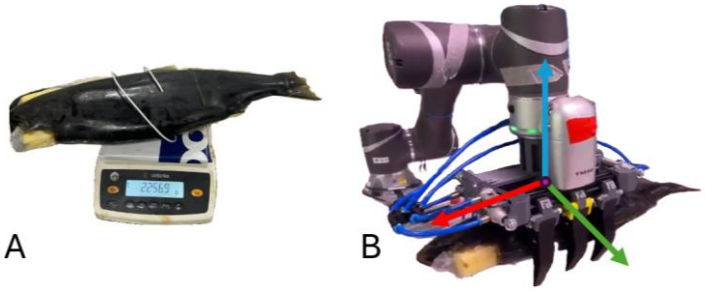


Figure 1: (A) Moulded fish with weight. (B) Annotations for the benchmarking, blue is vertical test direction, red is horizontal test direction.

Prototyping Gripper Designs

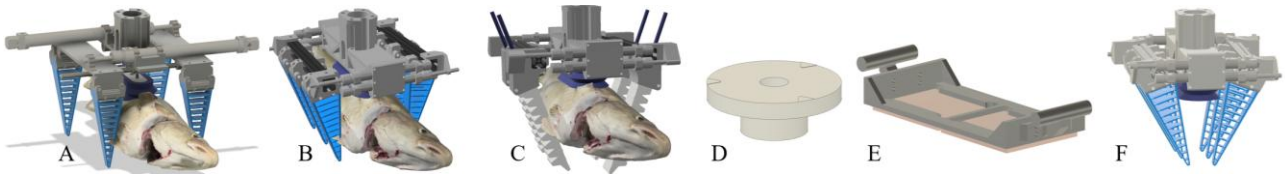


Figure 2: (A) First design of compliant finger and suction cup. (B) Second design of compliant finger and suction cup. (C) Silicone finger and suction cup design. (D) Bernoulli gripper design. (E) Freeze gripper design. (F) Angled compliant fingers and suction cup.

Compliant fingers and suction cup

The first gripper concept attempted to combine multiple compliant fingers together with a suction cup. The idea was to split the grasping problem into two parts, where the suction cup would account for the vertical lift force. Afterwards the fingers were actuated to prevent the possible movement of the fish in the vacuum cup. The first design iteration of the gripper is shown in Figure 2A. In this design, the suction cup was mounted in the middle of the gripper with springs to adapt to different sizes of fish. The compliant fingers were mounted on guided plates that were actuated by pneumatic cylinders. This way the fingers could adapt to different widths. The final part of the design is an adapter between the suction cup plate and the flange of the TM5 robotic arm.

The compliant finger and suction cup design was then redesigned to make it more scalable and robust (see Figure 2B). In the second iteration, the pneumatic cylinders were changed for smaller ones and uses two cylinders for each movement. The fingers were mounted on aluminium extrusions which makes it possible to add or remove fingers as seen fit. It also enabled using the frame of this design to test other combinations. The adapter between the suction cup plate and the robot arm was combined into the gripper frame with some added parts for overall robustness.

After several iterations and benchmark tests (prototypes A-E), another iteration was conducted on the compliant finger and suction cup design. Here, one of the fingers was removed to enable folding, and the remaining fingers were angled 10 degrees towards the suction cup to make the fingers overlap and wrap around the fish more. See Figure 2F. Additionally, this could allow the fingers to scoop under the fish when initiating the grasp.

Silicone fingers and suction cup

In the second gripper concept, the same base designed for the suction cup and compliant finger gripper was used. However, compliant fingers were swapped for flexible silicone fingers with internal pneumatic chambers. The flexible fingers would function similarly to the compliant fingers, but the main difference was that the fingers can be moved away from the fish by applying a vacuum. This way the fingers could avoid colliding with other fish when approaching the fish to pick. Figure 2C shows this gripper design.

Bernoulli suction

Bernoulli suction failed only at the uneven surface requirement in Table 3. It was therefore decided to first test this technology isolated to investigate shortcoming with respect to uneven surfaces, so that another solution could be added in combination. The design was heavily influenced by the principal drawings and graphs from Dini et al., (2009). Here, the gripping force (F_g) of a Bernoulli gripper is calculated as:

$$F_g = F_l - F_r \ (1)$$

$$F_l = \frac{1}{2}\rho\frac{Q^2}{2\pi h^2}\left[ln\frac{r_{ext}}{r_{int}} - \frac{1}{2}\left(\frac{{r_{ext}}^2-{r_{int}}^2}{{r_{ext}}^2}\right)\right] \ (2)$$

$$F_r = \rho\frac{Q^2}{\pi {r_{int}}^2} \ (3)$$

Where F_g is total grip force, F_l is the lift force, F_r is repulsion force, ρ is ambient air density, Q is air flow, h is height if the air gap, r_{ext} and r_{int} is external and internal radius of the gripper (Dini et al., 2009).

The first design of the Bernoulli gripper prototype can be seen in Figure 2D. In this design the internal radius was 0.005 m, the external radius is 0.02 m, and the height offset 0.0002 m. Using a flow rate calculator, the air flow from the compressor was estimated to be approximately 0.02 m^3/s, and the air density at room temperature was estimated to be 1.2 kg/m^3. With these parameters, the grip force can be calculated as:

$$F_l = \frac{1}{2}1.2\frac{0.02^2}{2\pi 0.0002^2}\left[ln\frac{0.02}{0.005} - \frac{1}{2}\left(\frac{0.02^2 - 0.005^2}{0.02^2}\right)\right] \approx \mathbf{876\ N}\ (4)$$

$$F_r = 1.2\frac{0.02^2}{\pi 0.005^2} \approx \mathbf{611\ N}\ (5)$$

$$F_g = 876N - 611N = 265\ N \approx \mathbf{26.5\ kg}\ (6)$$

The calculation included rough estimations and therefore needed benchmarking before combined with other solutions.

Freeze gripper

The first prototype of the freeze gripper aimed to benchmark the gripping capability of two 85x90 mm copper plates. This way, the plates could be mounted in a bigger grid in future prototypes to lift heavier loads. The design is shown in Figure 2E.

By dipping the moulded fish in water, and applying the freezing gripper contact area, a thin ice layer can be created. Experiments described by Terrile et al., (2021) concluded that the holding force of the ice layer is larger than the rupture force in fish fillets. Additional tests on textiles showed that freezing 10-30% of the grippers contact area resulted in a rupture force of 9-15 N/cm^2. The benchmarking needs to measure the contact area and holding forces to investigate if the technology is suitable for whole fish picking.

Results

This section presents the benchmark test from each prototype design and summarize the findings to indicate which design holds most promise.

Compliant fingers and suction cup

The initial iteration of the compliant finger and suction cup design, and the respective benchmark test is shown in Figure 3A. Through the benchmarking, the vertical and horizontal holding capacities were found to be 92.5 N and 15 N.

The second iteration and the benchmark test can be seen in Figure 3B. In this iteration the vertical and horizontal holding capacities were found to be 135 N and 35 N.

The third design iteration and benchmark test is shown in Figure 3F. This prototype had a substantial increase in vertical holding capacity which was measured to be 175 N, but the horizontal holding capacity stayed at 35N.

An additional test was conducted where the vacuum system for the suction cup was replaced by an industrial vacuum pump. Here the vacuum was measured to be 50 mbar, and resulted in a measured vertical holding capacity of 215 N and a horizontal holding capacity of 65 N.

Silicone fingers and suction cup

The prototype of the soft robotics finger and suction cup gripper, and the benchmark can be seen in Figure 3C. This grippers benchmark resulted in a vertical and horizontal capacity of 87.5 N and 20 N. An additional test was conducted where only the suction cup was actuated to find how much vertical lift the fingers added. The suction cups vertical and horizontal capacity was measured at 82.5 N and 15 N.

Bernoulli suction

The Bernoulli gripper prototype, and the corresponding benchmark is shown in Figure 3D. The grasp force was substantially lower than the theoretical estimate, failing to pick the moulded fish at all. The lift force was found by picking up a cardboard box and filling it with added weight until the grasp failed. The grasp force was measured to be around 7 N. During the benchmarking, it was observed that the airflow from the compressor would drop almost immediately when actuating the gripper. Equation (2) can be changed to solve for airflow, which gives an approximate airflow of 0.0018 m^3/s or 108 l/min at 7 N grasp force.

Freeze gipper

The freeze gripper prototype was simplified to only use one of the copper plates as the moulded fish was not deformable enough to get good contact with both plates. As only one plate was used, it was easier to freeze the plate by remove the gripper frame and place it on the fish by hand. Dry ice at about $-78.5℃$ was used to efficiently cool down the copper plate. This led to the water layer between the moulded fish and the copper plate to freeze seemingly instantly. The measured

vertical and horizontal holding force was measured to be 50 N, and the contact area was measured to be 45x90 mm, meaning that only 53% if the surface area was utilized. The freeze gripper prototype, and benchmark is shown in Figure 3E.

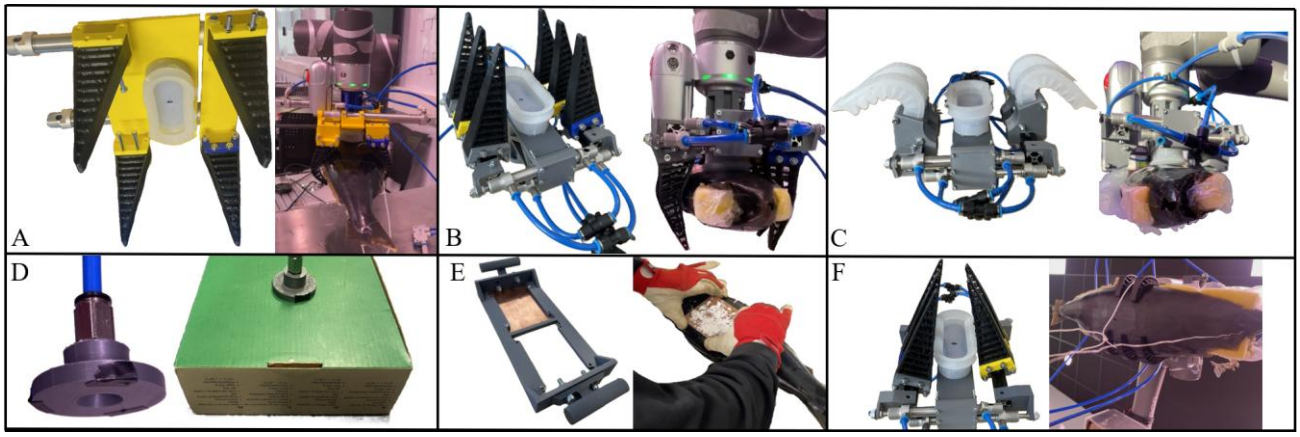


Figure 3: (A) First iteration of compliant finger and suction cup, and snippet from benchmarking. (B) Second iteration of compliant finger and suction cup, and snippet from benchmarking. (C) Prototype of silicone finger gripper and snippet from benchmarking. (D) Prototype of Bernoulli gripper and snippet from benchmarking. (E) Freeze gripper prototype and snippet from benchmarking. (F) Third iteration of compliant finger and suction cup, and snippet from benchmarking.

Comparisons of gripper benchmarks

Figure 4 shows a plot from each prototype benchmark. The blue bars represent the measured vertical and horizontal grasp forces on isolated technologies, the orange bars represent the grasp forces on combined technologies, and the red line shows the edge-case requirement from the Baader 444 specifications. The graph shows that prototypes B and F (Figure 3B and 3F) which combines a suction cup and compliant fingers are the only designs that can handle the edge-case requirement.

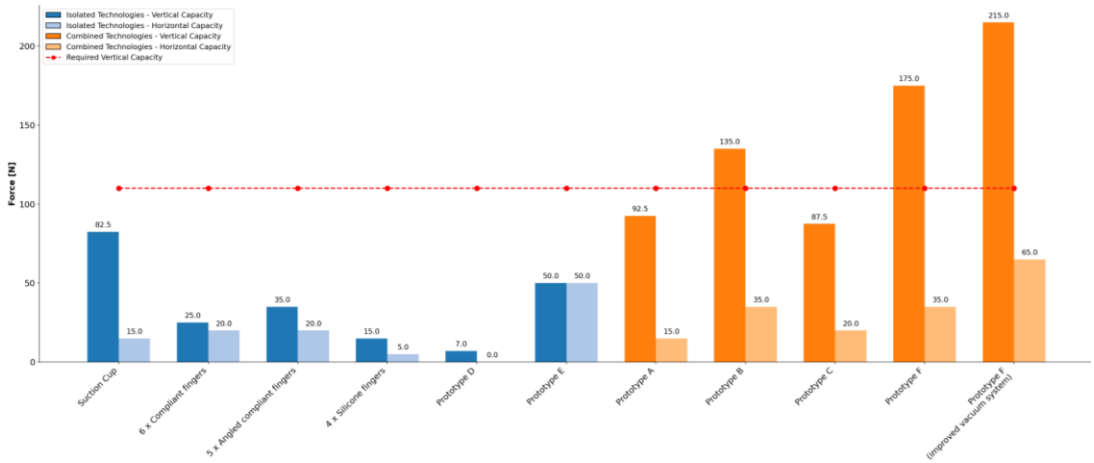


Figure 4: Comparisons between benchmark results of the gripper prototypes and technologies.

Discussion & Conclusions

Through benchmark tests of robot gripper prototypes with respect to load capacity, it was found that combining a suction cup with compliant fingers yielded the most promising results. It shall be noted that this prototype was also developed to highest fidelity. Therefore, a comparison between this prototype and the isolated technologies should be made with caution. To find the optimal combination, it would be useful to further develop and benchmark different combinations of gripping technologies including the ones not tested from the morphology table. However, some conclusions can be drawn.

Looking at the isolated technologies, the compliant- and silicone fingers had a low vertical capacity without the help of the suction cup. This contrasts findings from Dyrstad et al., (2018); Dyrstad and Mathiassen, (2017); Fang et al., (2023); Misimi et al., (2018), where about 75% of the fish was grasped successfully. This indicates that further improvements on the compliant fingers would likely improve the overall load capacity when used in a hybrid configuration. It was also clear that the suction cup was far superior to the Bernoulli gripper and would be preferred in a hybrid configuration. From Table

3, the freeze gripper concept looked promising for most requirements. However, the load capacity of about 50 N was less than the suction cup. It is likely that increasing the contact area would increase the load capacity, e.g. by making the copper plate flexible. It shall be noted however that the low temperature of $-78.5℃$ would likely damage the fish and impair food quality. Therefore, the effect and threshold of freezing gripper temperatures should be studied in detail.

The dummy fish model used for the benchmark experiments was based on a polyurethane-sponge based design that was coated with soap. A limitation with this model was that it does not fully reflect the biomechanical behaviour of real fish. This is of particular interest concerning fish shells in combination with suction cups, and the results presented in this study are likely not conservative. Conducting the benchmark experiments on real fish will provide better insight into industrial limitations and is planned for near future research. Also, there is currently no consensus on the minimum allowable grasping force or pressure that causes damage to the fish, and should be investigated in future research.

The vertical lift capacity requirement from the Baader 444 machine, as highlighted in Figure 4, was based on static lifting. By introducing dynamic behaviour, this holding capacity requirement would increase. Assuming that three robot arms feed 35 fish per minute at 11 kg each to the machine, moving a vertical distance of 0.2 m, a horizontal distance of 1 m, and symmetric constant acceleration/deceleration profile, we can calculate theoretical dynamic forces. We assume the cycle to be grasping (0.9 sec), lift (0.6 sec), move (x sec), lower (0.4 sec), release (1 sec), lift (0.6 sec), move (x sec) and lower (0.4 sec). Vertically, the robot end-effector would need to lift 0.1 meters in 0.3 seconds, resulting in an acceleration of $2.2\ m/s^2$ and a force of 132 N including gravity. Horizontally, the robot end-effector would need to move 0.5 meters in 0.44 seconds, giving an acceleration of $5.3\ m/s^2$, and a lateral force of 58 N. Prototype F still meets these requirements. It shall be noted that the lateral horizontal force capacity was measured in the worst-case orientation (see Figure 1B), and it is likely that a much higher horizontal capacity can be achieved with the presented hybrid designs. This demonstrates industrial implementation feasibility.

We hypothesized that multiple grasping technologies had to be combined to meet industrial requirements for fish grasping and load capacity. The benchmark tests supported this assumption, where prototypes B and F, which integrated both suction cup and compliant finger technologies, were the only designs that fulfilled the specified performance criteria. Therefore, we could not falsify our hypothesis. It remains possible that a single grasping technology could be sufficient if carefully optimized for a narrowly defined application or if the variability in fish size, shape, and behavior were reduced.

From a 'design-for-AI' perspective, an important ingredient is how hardware morphology offloads computational complexity from the control algorithms. Controlling gripping force to avoid product damage is already notoriously challenging, as fish exhibit nonlinear material behaviour and friction. Soft or compliant robotic gripper designs that passively conforms to the fish shape can effectively offload this complex grasp control to the hardware. An alternative design approach would be rigid articulated robotic hands, such as those found in humanoids, but this instead pushes complexity into expensive hardware and sophisticated control, and deformable object manipulation remains a key unsolved problem. Additionally, for fishing trawler vessels operating in remote locations, internet connectivity is limited and compute limitations make lightweight edge-AI a necessity. This is important also because the dynamic marine environment introduce operational complexity in terms of positional perception and robotic path planning, which require much of the available onboard computing power.

Because physical AI perception and control methods typically require large training datasets, simulation-driven design is attractive due to efficient generation of synthetic data. Inherently, there exist a bias toward also moving hardware design activities into virtual prototyping as well, since hardware and control are tightly coupled. This contrasts the prototype-heavy design strategy employed in this study. While beyond the scope of this paper, related efforts have addressed capturing data for improving the simulation environment (Karlsen et al., 2026). Collectively, we argue that the persistent sim-to-real gap, that arises from the inability of simulators to fully capture real-world contact, compliance, and nonlinear material behaviour, remains a fundamental bottleneck not only for control but for hardware design, and that physical prototyping remains necessary in addition to simulation activities.

Acknowledgement

The authors would like to acknowledge Irina-Emily Hansen, Ola Jon Mork, Andreas Flem Norman, Ishfaq Bhat and Magnus Mortensen at NTNU for fruitful conversations about this topic. We thank the Norwegian Seafood Research Fund for funding the project (project 901991 SIM-FISHPROCESSING).

References

Caldwell, D.G., 2013. Robotics and automation in the food industry: current and future technologies, 1st edition. ed, Woodhead Publishing series in food science, technology and nutrition, number 236. Woodhead Publishing Limited, Cambridge, UK.

Danielsen, A., Winsjansen, R., 2019. Dexterous Gripper: modular platform for robotic manipulations (Master Thesis). Norwegian University of Science and Technology.

Dini, G., Fantoni, G., Failli, F., 2009. Grasping leather plies by Bernoulli grippers. CIRP Annals 58, 21–24. https://doi.org/10.1016/j.cirp.2009.03.076

Dyrstad, J.S., Mathiassen, J.R., 2017. Grasping virtual fish: A step towards robotic deep learning from demonstration in virtual reality, in: 2017 IEEE International Conference on Robotics and Biomimetics (ROBIO). Presented at the 2017 IEEE International Conference on Robotics and Biomimetics (ROBIO), pp. 1181–1187. https://doi.org/10.1109/ROBIO.2017.8324578

Dyrstad, J.S., Ruud Øye, E., Stahl, A., Reidar Mathiassen, J., 2018. Teaching a Robot to Grasp Real Fish by Imitation Learning from a Human Supervisor in Virtual Reality, in: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). Presented at the 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 7185–7192. https://doi.org/10.1109/IROS.2018.8593954

Espelund, J., 2022. Robotisert handlegging av makrell i 10 kilos kasser. Optimar AS.

Fang, H.-S., Wang, C., Fang, H., Gou, M., Liu, J., Yan, H., Liu, W., Xie, Y., Lu, C., 2023. AnyGrasp: Robust and Efficient Grasp Perception in Spatial and Temporal Domains. IEEE Transactions on Robotics 39, 3929–3945. https://doi.org/10.1109/TRO.2023.3281153

Giske, L., Espelund, J., Flem, A., 2025a. Konsept for helhetlig automatisert havfiskefartøy. Optimar.

Giske, L., Espelund, J., Flem, A., 2025b. Singulering. Optimar.

Hatlem, T., 2026. Prisvekst på villfisk og økt laksevolum ga verdirekord for sjømateksporten i 2025 [WWW Document]. Fisk.no. URL https://fisk.no/fiskeri/7901-prisvekst-pa-villfisk-og-okt-laksevolum-ga-verdirekord-for-sjomateksporten-i-2025 (accessed 2.14.26).

Karlsen, B., Urcelay, J., Mortensen, M., Mork, O.J., Hansen, I.E., Kleppe, P.S., Norman, A., Giske, L.A., Bjelland, O., 2026. Iterative Industrial Prototyping for Industry 4.0: Data- Collection Systems for Digital Twins in Marine Seafood Processing, in: Proceedings of the Design Society, Volume 6: DESIGN 2026. Presented at the DESIGN 2026, Design Society, Dubrovnik.

Khan, Z.H., Khalid, A., Iqbal, J., 2018. Towards realizing robotic potential in future intelligent food manufacturing systems. Innovative Food Science & Emerging Technologies 48, 11–24. https://doi.org/10.1016/j.ifset.2018.05.011

Kvam, D., Flem, A., Espelund, J., 2023. Automatisk pakkebord. Optimar AS.

Low, J.H., Khin, P.M., Han, Q.Q., Yao, H., Teoh, Y.S., Zeng, Y., Li, S., Liu, J., Liu, Z., Valdivia y Alvarado, P., Chen, I.-M., Tee, B.C.K., Yeow, R.C.H., 2022. Sensorized Reconfigurable Soft Robotic Gripper System for Automated Food Handling. IEEE/ASME Transactions on Mechatronics 27, 3232–3243. https://doi.org/10.1109/TMECH.2021.3110277

Misimi, E., Olofsson, A., Eilertsen, A., Øye, E.R., Mathiassen, J.R., 2018. Robotic Handling of Compliant Food Objects by Robust Learning from Demonstration, in: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). Presented at the 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 6972–6979. https://doi.org/10.1109/IROS.2018.8594368

Phan, K., 2025. Robotic Bin-Picking of Raw Salmon Fillets Using Deep Instance Segmentation and Stereo Vision. https://doi.org/10.2139/ssrn.5517081

Reite, L.B., 2023. Design av konsept for prototype av smart robotgriper for fisk (Bachelor Thesis). Norwegian University of Science and Technology, Ålesund.

Steinert, M., Leifer, L., 2012. "Finding One's Way": Re-Discovering a Hunter-Gatherer Model based on Wayfaring. International Journal of Engineering Education 28, 251–252.

Terrile, S., Argüelles, M., Barrientos, A., 2021. Comparison of Different Technologies for Soft Robotics Grippers. Sensors 21, 3253. https://doi.org/10.3390/s21093253

Wang, Z., Furuta, H., Hirai, S., Kawamura, S., 2021. A Scooping-Binding Robotic Gripper for Handling Various Food Products. Front. Robot. AI 8. https://doi.org/10.3389/frobt.2021.640805